

Day-03

DOCUMENTATION

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Objective:

The objective of learning Transformers is to understand how AI models process text, identify relationships between words, and generate meaningful responses. Transformers use the Attention Mechanism to understand context and form the foundation of modern language models like ChatGPT, Gemini, and BERT.

Transformers:

1. What are Transformers?

Transformers are deep learning architectures used in modern language models. They process text efficiently and understand relationships between words using the Attention Mechanism.

2. Why were Transformers introduced?

Before Transformers, RNNs and LSTMs were used for sequence processing. They struggled with long-range dependencies and remembering information from earlier parts of long sentences. Transformers solve this problem by processing all words simultaneously.

3. Basic Architecture:

Input Text

Tokenizer

Embeddings

Transformer Layers

Output

4. Role of Attention Mechanism:

Attention helps the model focus on important words and

understand relationships between them.

Example:

"Ram gave a book to Sita because she needed it."

The Attention Mechanism helps identify that "she" refers to "Sita".

- **DATA COLLECTION FOR THE PROPOSED SLM**

Collected Data Sources & Why These Data Sources are Useful:

1. **NCERT Textbooks**

Type: PDF / Educational Content

NCERT textbooks contain detailed explanations of school subjects such as Mathematics, Science, Social Studies, and English. These books provide accurate academic content, examples, and exercises.

Why Useful for the SLM:

These textbooks help the Education SLM answer subject-related questions, explain concepts clearly, and support students in their learning process.

2. **Khan Academy**

Type: Website

Khan Academy provides free educational lessons, tutorials, videos, quizzes, and practice exercises for various subjects.

Why Useful for the SLM:

This resource helps the SLM provide step-by-step explanations and improve its ability to guide students through different topics.

3. **Wikipedia Educational Articles**

Type: Website

Wikipedia contains educational articles on science, mathematics, history, geography, technology, and many other subjects.

Why Useful for the SLM:

It provides general knowledge and educational information that can help the SLM answer a wide variety of student questions.

4. **SQuAD Dataset**

Type: Dataset

The Stanford Question Answering Dataset (SQuAD) contains passages along with questions and answers based on those passages.

Why Useful for the SLM:

This dataset helps the SLM learn how to understand educational content and provide accurate answers to student queries.

5. OpenStax Textbooks

Type: Website / Educational Content

OpenStax provides free and high-quality textbooks for various academic subjects.

Why Useful for the SLM:

These textbooks offer reliable educational content that can improve the knowledge base of the proposed Education SLM.

Links

Websites:

- <https://ncert.nic.in/>
- <https://www.khanacademy.org/>
- <https://www.wikipedia.org/>
- <https://www.wikipedia.org/>

Datasets:

- <https://raipurkar.github.io/SQuAD-explorer/>
- <https://huggingface.co/datasets>